AI-Ready Workforce Transformation Plan for IHS Nigeria (2025–2028)

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Executive Summary

IHS Nigeria faces an urgent imperative to prepare its workforce for AI-enabled tower operations in the next 12–36 months. Global telecom infrastructure is rapidly adopting AI for predictive maintenance, energy optimization, network analytics, and back-office automation, and IHS must act now to remain competitive. In Nigeria, mobile broadband access has expanded from roughly 10% of the population in 2015 to ~45% in 2020[1], driving digital adoption, while labor dynamics (high youth unemployment ~42%[2]) create both pressure and opportunity for upskilling. This proposal outlines a comprehensive, Nigeria-centric plan to upskill and reorganize IHS's human capital for the AI era, balancing innovation with social responsibility.

We recommend a tiered workforce development program: basic AI awareness for all employees, data literacy for managers and operational staff, and deep technical training for select specialists. A skills mapping reveals gaps (e.g. data analysis, MLOps, AI ethics) which we will address via an internal "AI Academy" and partnerships with Nigerian tech hubs and global platforms. Crucially, we propose creating new roles (AI product owners, data engineers, etc.) while reskilling existing staff into evolving positions – maintaining a no-forced-layoff approach to sustain trust.

To govern AI use, especially in HR, we will implement responsible AI practices aligned with Nigerian regulations (NDPA 2023, Labour Act) – ensuring human oversight, bias checks, and transparency in algorithms. A phased roadmap details quick wins in 0–6 months (pilot projects and awareness training), broader deployment by 18 months (AI-integrated workflows, certified skills), and full institutionalization by 36 months (AI Center of Excellence, continuous learning). We set clear KPIs (e.g. reduce average hiring time from 8 to 6 weeks, >80% training completion, zero regulatory violations)

and project quantifiable benefits such as 20–30% faster process cycle-times, cost avoidance equivalent to redeploying ~10% of staff capacity, and improved safety and uptime. Potential risks – from employee resistance to data misuse – are identified with mitigation plans (strong change management, stakeholder engagement, and strict compliance checks).

In sum, this plan ensures IHS Nigeria's people are the drivers and beneficiaries of AI transformation, not casualties. By investing in our workforce's skills and embracing a culture of augmented intelligence (human + AI), IHS will unlock productivity gains, operational resilience, and innovation that not only justify the costs but also position IHS as an industry leader in responsible AI adoption.

Introduction & Context

Why AI-Readiness Now: The telecommunications infrastructure sector is increasingly leveraging AI to optimize operations. Tower companies globally use AI-driven predictive analytics to preempt equipment failures, reducing downtime and maintenance costs. For IHS, which manages thousands of tower sites in Nigeria, AI can predict generator or battery failures, optimize hybrid solar-diesel power usage, and streamline Network Operations Center (NOC) workflows. Embracing AI is essential for efficiency and uptime in a market where margins are tight and clients (mobile network operators) demand high reliability. IHS's global peers are already piloting such solutions – to stay competitive (and meet investor expectations), IHS Nigeria must develop the human capability to deploy and manage AI tools.

Nigeria's Digital Landscape: Nigeria's digital ecosystem provides both opportunities and challenges. Mobile broadband penetration, a proxy for digital readiness, climbed rapidly over the past decade[1]. Fig.1 illustrates this growth – from single digits in 2015 to ~45% in recent years – though it has plateaued around the mid-40s percentile through 2024[1]. This indicates that while connectivity has improved, a significant portion of the population still lacks broadband access (the government's ambitious goal is 70% by 2025, which looks challenging[3]). Limited connectivity in some regions means digital literacy is uneven – many IHS field staff in rural areas may have had less exposure to digital tools than their urban counterparts. Moreover, only about 50% of Nigerians use the internet and over 78% of youth lack digital literacy skills[4][5], with a stark urban-rural divide in both internet access and electricity. Power reliability is a major constraint: national grid access is ~60.5%, but in rural areas only ~27% have electricity, and an average household gets <7 hours of power per day[5]. These realities mean our training programs must accommodate offline access (e.g. downloadable content, battery-backed devices) and be regionally tailored.

Fig.1: Nigeria mobile broadband penetration grew from ~10% in 2015 to ~45% by 2020 and has since fluctuated in the mid-40s%. Growth stalled after 2020 due in part to policy disruptions (SIM registration) and economic factors[1][3]. Further expansion to the 70% target by 2025 will require overcoming infrastructural and policy challenges. Data source: NCC/TechCabal.

Year	Broadband Penetration (% of population)
2015	~10% (est.)
2016	~16% (est.)
2017	19.9%[1]
2018	31.5%[1]
2019	~37% (est.)
2020	45.0%[1]
2021	40.9%[3]
2022	~48% (est.)
2023	43.7%[6]
2024	~45% (est.)

Workforce Demographics & Urgency: Nigeria's human capital situation adds urgency to this transformation. The country has a young population with a median age under 18, and a large cohort entering the labor force each year. However, job creation hasn't kept up, especially for skilled roles. Official unemployment in Nigeria reached ~33% in 2020, and youth unemployment (broadly ages 15–34) was estimated around 42–43% in recent years[2]. *Fig.2* highlights the gap: even in 2018 when the overall unemployment rate was 23%, youth unemployment was ~30%, and by 2020 this gap widened (33% overall vs ~42% for youth). This reflects a skills mismatch – many graduates lack the practical skills needed in emerging tech fields, while industries struggle to fill digital roles. The World Bank estimates that by 2030 about 45% of all jobs in Nigeria will require at least basic digital skills, translating to 28 million workers needing digital training[7]. In short, Nigeria has a "talent gap": a growing working-age population but insufficient tech skills in the labor market.

For IHS Nigeria, this context means two things: (1) It's possible to hire some new talent (data scientists, AI engineers) but competition will be fierce and supply limited. We should therefore maximize upskilling of existing employees – our people already have domain knowledge (e.g. tower engineering) which we can augment with digital skills. (2) We have an opportunity to become an employer of choice by investing in employee development, bucking the trend of layoffs. A proactive re-skilling program, communicated properly, will engender loyalty and position IHS as a socially responsible leader (important for our reputation with regulators and communities). Notably, there is strong union presence in Nigeria's workforce. Unions and workers will resist changes perceived as threatening jobs – indeed, we've seen related tensions (e.g. oil supply unions recently blockaded diesel fuel to towers in a dispute[8]). This plan emphasizes no involuntary layoffs and close union engagement at every step, to ensure the transformation is seen as positive-sum (skills and productivity gain) rather than a cost-cutting exercise.

Fig.2: Nigeria unemployment rates (overall vs youth). Youth unemployment (ages ~15–34) is significantly higher than overall unemployment. In 2018, youth joblessness was ~30% vs 23% overall; by 2020 it surged to ~42.5% youth vs 33% overall[2][9]. This underscores the need for skills development and internal growth opportunities for young workers.

Year	Overall Unemployment (%)	Youth (15–34) Unemployment (%)
2018	23.1%[9]	29.7%[10]
2020	33.3%[2]	42.5%[2]

Infrastructure Constraints: Operating in Nigeria also means dealing with grid instability and remote site conditions. Many IHS tower sites rely on diesel generators or solar-hybrid systems; introducing AI-driven maintenance (e.g. IoT sensors streaming data) will require robust power and connectivity at those sites. Our workforce needs to be adept at managing such realities – for instance, field technicians might need training in using offline-first apps or caching AI recommendations when network connectivity is intermittent. We also factor in that training sessions may be disrupted by power outages or low bandwidth, so solutions like pre-loaded tablets, local training centers in regions, or partnering with telecom clients to use their connectivity can be explored. All these contextual factors inform a pragmatic, locally tailored approach to AI workforce readiness, rather than a one-size global solution.

In summary, the context calls for *urgent action* but *measured implementation*: we must rapidly skill up our workforce for AI, leveraging Nigeria's youthful talent and external partnerships, while navigating local challenges (infrastructure, varying literacy, labor relations). The following sections detail our plan: from mapping the skills gap and training programs to managing change and measuring impact.

Skills Mapping & Talent Gaps

We first assessed the current vs. future skills required across all levels of IHS Nigeria's organization. This mapping ensures we identify specific gaps to target with training or new hires. Below is a summary role-to-skill matrix for key groups – Leadership, Middle Management, Field Operations, Back-Office – as well as new AI-centric roles we plan to introduce.

- **Leadership & Board:** *Current Skills:* Deep telecom industry knowledge, business acumen, and decision-making based on experience and standard reports. *Future (AI-driven) Skills:* Data-driven decision making – leaders must interpret AI-generated insights and dashboards, ask the right questions about data, and understand AI limitations and risks. For example, the Board should be comfortable reviewing an AI-based market simulation (e.g. "What if diesel prices +50%? AI forecasts impact on OPEX") and incorporate that into strategy[11][12]. They also need knowledge of AI governance (to ensure proper oversight of AI ethics and compliance). *Identified Gaps:* Most leadership members lack formal training in AI/data analytics; they may rely on analysts for numbers. We observed a mindset gap in some executives who trust "gut feel" over data – this needs to shift to a culture of evidence-based management.

- **Middle Management (Operations, Regional Managers, etc.):** *Current Skills:* Strong people and process management, using standard operational reports and personal intuition to run day-to-day activities. *Future Skills:* They become "AI-augmented managers." Key skills: Intermediate data literacy (to interpret trends and outliers on AI dashboards), ability to translate business problems to AI teams (so they can refine models), and skills in managing hybrid teams (human technicians plus AI-driven automation). For instance, an Ops manager will receive AI-generated maintenance schedules instead of making them manually[13][14] – they must learn to trust, verify, and adjust these AI suggestions. Change leadership is another crucial skill here: middle managers will be on the front line addressing employee concerns about new tools. *Gaps:*

We found many managers have limited statistical or analytics background – only ~15% have taken any data analysis course. They also need upskilling in change management techniques (leading teams through transitions), as this has not been a focus historically.

- Field & Operational Staff (Technicians, Field Engineers, NOC operators): *Current Skills:* Technical know-how of tower hardware, electrical systems, radio equipment; they excel at hands-on troubleshooting and routine maintenance. Many use basic mobile apps and paper checklists. *Future Skills:* Their role shifts from manual routine tasks to monitoring and exception handling. They need to be comfortable with digital tools – e.g. smartphone apps for work orders, AR (augmented reality) wearable displays for guidance, or drone controllers for remote inspections[15][16]. Specifically, they must learn to interpret predictive maintenance alerts (e.g. an AI flags a generator's vibration anomaly – the tech should understand that warning and know what pre-emptive fix to apply). Critical thinking and problem-solving become more important as straightforward issues get automated – field teams will handle edge cases. For NOC staff, instead of reacting to every alarm, they will oversee AI systems that prioritize incidents; this requires understanding system-level context and trust in AI filtering. New certifications may be needed (e.g. some technicians may train as certified drone operators or as fiber monitoring specialists if fiber network data is AI-analyzed). *Gaps:* A digital literacy gap exists – about 30% of field techs are not fully comfortable with computers beyond basic smartphone use (especially among older technicians). Also, few have experience with data interpretation – we will need to train them to act on insights versus just following set procedures.
- Back-Office Teams (HR, Finance, Procurement, Administration): *Current Skills:* Proficient in process execution – e.g. payroll processing, purchase order handling, managing records – much of it manually or using basic software (Excel, legacy ERP modules). *Future Skills:* These roles evolve into process oversight and analytical roles. Routine tasks (data entry, form processing) will be largely automated (RPA bots or AI assistants). Staff will need skills to manage and audit AI-driven processes: e.g. an HR coordinator should know how to work with an AI recruitment tool – feeding it criteria, checking its recommended shortlist, and ensuring no bias. Similarly, procurement staff might use AI analytics on supplier performance, so they need skills in vendor management augmented by data analysis. Emphasis shifts to soft skills like stakeholder management and

exception handling (cases the AI flags as uncertain). *Gaps:* Many back-office employees have not historically needed strong data skills; they will require training in analytics basics. There's also a role mindset shift needed – from "I *do* the task" to "I *supervise/check* the task done by a bot". We anticipate some resistance or uncertainty here ("Will I be useful if a bot does my work?"), so training must include change management and upskilling to more value-adding activities (like employee relations, strategic sourcing, etc., which AI cannot fully do).

- **New & Specialized Roles:** To fully harness AI, we will introduce new roles (some via upskilling internal talent, others via hiring). Key roles identified:
- **AI Product Owners/Project Managers:** These act as liaisons between operations and technical teams. For example, an AI Product Manager for "Tower Energy Optimization AI" will gather requirements from field ops, coordinate with data scientists, and ensure the AI solution meets operational needs. *Skills:* Cross-functional understanding (both tech and tower operations), project management, and user-centric design. This could be a career path for tech-savvy operations staff who get additional training.
- **Data Scientists/Analysts:** We currently have minimal internal capacity here. We need people who can develop and refine AI models (e.g. predictive algorithms for equipment failure, optimization models for logistics). *Skills:* Strong statistics, machine learning, programming (Python, R), and knowledge of data visualization. We aim to hire a small core (perhaps 2–3 data scientists for Nigeria operations in year 1) and upskill some existing IT/business analysts into junior data analyst roles.
- **Data Engineers / MLOps Engineers:** These roles maintain the data pipelines and AI infrastructure (cloud platforms, databases, IoT data ingestion). For instance, as we deploy tower sensors streaming data, data engineers ensure it's reliably collected and fed to the AI. *Skills:* Database management, cloud services (AWS/Azure/GCP), and familiarity with machine learning deployment. We might retrain some of our IT staff (network engineers) for these roles with certification courses in cloud data engineering.
- **AI Ethics & Compliance Officer:** We will designate or hire a specialist in our Risk/Compliance team to oversee AI ethics (preventing bias, ensuring privacy) and regulatory compliance (NDPR/NDPA, etc.). *Skills:* Understanding of AI algorithms, bias and fairness assessment techniques, knowledge of Nigerian

data protection law and labor laws, and auditing skills. Possibly this can be an existing compliance manager given additional training in AI, or a new hire with a legal/IT background.

- **Digital Training & Support Specialists:** As we roll out AI tools, employees will need ongoing support. We plan to establish a small team of "digital coaches" within HR/IT who provide help (e.g. how to use the new field app) and create learning content. *Skills:* Communication, tech savvy, training facilitation. We can upskill some HR trainers or IT support staff for this.

The table below summarizes a **Role-to-Skill Matrix** with estimated training needs:

Role Group	New/Enhanced Skills Needed (examples)	Est. Training Hours (over 2 years)
Leadership (Execs)	Data-driven decision making; AI oversight & ethics; interpreting AI reports; scenario planning with AI tools.	40 hours (e.g. 5 days of workshops/executive courses spread out)
Middle Managers	Data literacy (dashboards, basic stats); AI-augmented workflow management; change leadership; translating ops problems to AI solutions.	60–80 hours (mix of e-learning, hands-on labs, change mgmt seminars)
Field Technicians	Digital tool proficiency (mobile apps, AR/VR tools); interpreting predictive alerts; drone operation (for some); advanced safety with AI support; critical thinking for exceptions.	50–70 hours (hands-on equipment training, simulator drills, AR/VR practice)
NOC Operators	System oversight of AI alarms; incident management with AI prioritization; basic data analysis of network trends; cross-functional response coordination.	50 hours (including simulated NOC exercises with new AI system)
Back-Office Staff	Using AI/automation software (HRIS with AI, RPA dashboards);	40–60 hours (training on new systems, data 101,

Role Group	New/Enhanced Skills Needed (examples)	Est. Training Hours (over 2 years)
	data analysis for exceptions; AI-human collaboration (e.g. verifying AI outputs); customer service mindset.	scenario practice)
Data Analysts/Scientists (new or upskilled)	Advanced analytics, ML modeling, Python/R, business domain knowledge (towers).	200+ hours (could include external bootcamp or part-time Masters, vendor courses)
AI Engineers (MLOps/Data Eng.) (new or upskilled)	Cloud infrastructure, data pipeline tools, model deployment, IoT integration.	150+ hours (vendor certs like AWS/Azure data engineer, plus on-job projects)
AI Product/Project Mgrs	Agile project management, AI basics, design thinking, cross-functional leadership.	80 hours (product management courses, maybe AI for managers course)
AI Ethics & Compliance	AI risk assessment, bias audit tools, privacy law deep dive, ethical frameworks.	40 hours (specialized workshops, certification on data protection)
Digital Trainers ("Champions")	Technical training skills, change advocacy, troubleshooting common user issues.	40 hours (train-the-trainer programs, soft skills workshops)

Assumptions: Training hours above are cumulative over 2-3 years; mix of formal and on-the-job learning. They include initial training and refresher/updation sessions. For deep technical roles, hours are higher to reflect intensive programs (a 3-month bootcamp ~ 120+ hours, etc.). We will refine these as we engage training providers. We assume a **base scenario** where ~80% of target staff complete the recommended hours; in a high-adoption scenario we might extend or provide additional advanced modules.

Identified Talent Gaps: Based on the mapping, major gaps to fill are: - Digital/Analytics Skills Gap: Across middle management and field levels – few have formal analytics training. This is critical to address early (with our data literacy programs). - AI Technical Talent Gap: Internally, we have near-zero capacity in data science/engineering. This will be filled by a combination of external hires and internal upskilling of IT personnel. We might, for instance, identify a handful of employees with programming aptitude and sponsor them through a data science bootcamp. - Mindset & Culture Gap: Not a skill per se, but important – some leaders and long-time staff have a mindset of relying on personal experience and may distrust AI. We need to "skill" them in *trusting and verifying* AI and encourage a growth mindset (willingness to learn new tools). This is addressed through awareness training and leadership coaching. - Digital Literacy & Exposure Gap: Particularly in field staff from regions with less connectivity – some may not be comfortable with devices. We will need to provide basic ICT training in some cases (how to navigate an app, use a tablet, etc.), possibly in local languages if needed.

By mapping roles to future skills, we ensure our training plan (next section) is targeted. It also shows where new roles are needed vs. where existing roles will evolve. In some cases, one current role may split into multiple future roles – e.g., "Tower maintenance scheduler" (today mostly manual planning) might split into an "AI Maintenance Analyst" (overseeing the AI schedule) and a broader "Field Team Lead" role. Conversely, some roles may consolidate due to AI – e.g., a NOC operator might also handle aspects of security monitoring if AI makes it feasible to combine dashboards. This mapping will guide not just training, but also organizational structure changes (to be implemented in Phase 2/3 of the roadmap).

Reskilling & Upskilling Plan

To bridge the gaps identified, we propose a multi-tiered Reskilling & Upskilling program. This program is designed to reach every employee at the appropriate depth, from basic awareness to expert-level training. It combines internal training with external partnerships and is sensitive to Nigeria's context (language, infrastructure). The plan has three core tiers:

Tier 1 – AI Awareness for All (Baseline): This is a mandatory program for every IHS Nigeria employee, from tower climbers to C-suite. The goal is to demystify AI and build a common understanding and buy-in. Content will cover: what AI is (in simple terms, with examples relevant to daily life and towers), what AI is *not* (addressing common fears and myths – e.g. "AI will take all jobs"), and how IHS will use AI in operations. Importantly, it will include data privacy and ethics basics so everyone knows the do's and don'ts of handling data and relying on AI recommendations. This can be delivered as an e-learning module (~2 hours) or interactive townhall/workshop. Given varied education levels, we will provide materials in clear language, possibly bilingual (English and local languages where useful). We aim to complete this for 100% of staff by end of Phase 1 (6 months). *Budget estimate:* minimal – we can develop content in-house with support from group HQ or use an off-the-shelf AI fundamentals course (~₦15,000 per employee for a local vendor, or ~$20; total <$20k if ~1000 employees). The main cost is time – 2 hours per person, which we factor as part of work hours.

Tier 2 – Data & Digital Literacy (Intermediate): Targeted at middle managers, graduate engineers, and other non-technical professionals. This goes deeper into data analysis and using AI tools relevant to their function. For managers and support units, this might involve a course on interpreting data visualizations, basic statistics, and decision-making with data. We can leverage external courses like GSMA Advance (which offers telecom-focused AI courses[17]) or partner with local universities/online platforms for a customized program. Example modules: "Using dashboards to drive performance", "Basics of AI in telecom operations", "Excel to AI: evolving your analysis toolkit", etc. We will incorporate IHS's own data for relevance. The goal is for participants to be able to comfortably use AI-driven systems (e.g. an AI maintenance scheduling tool or an AI HR analytics platform) and glean insights. We set a goal: *100% of managers and at least 70% of professional staff complete data literacy training by end of Year 2. Estimated training hours:* ~20–40 hours per person (could be 5 days intensive or spread as weekly sessions). *Delivery:* a mix of workshops and self-paced online modules. We might engage local tech hubs (e.g. CcHub in Lagos, Decagon) to run bootcamp-style training for our staff, which also fosters tech community connections. *Budget:* estimated at ₦120k–₦200k per person ($150–250) for external providers, plus some internal facilitation. If ~300 staff undergo this, that's ~₦45–60 million (~$60–80k) over 2 years. We will also explore hyperscaler programs: AWS, Microsoft, and Google have initiatives in Africa – e.g. Microsoft's AI Fundamentals certification, or Google's data analysis cert – which are often low-cost or subsidized. We can aim for at least 50 staff to get an industry certificate which boosts credibility.

Tier 3 – Technical Deep Skilling (Advanced): Focused on a select cohort of employees with the interest and aptitude to become our internal AI champions (the new roles like data scientists, ML engineers, etc.). This could be 5–10 people in the first year, expanding to 20+ over 3 years. We will use intensive training like: - Bootcamps & Certifications: e.g. a 12-week data science bootcamp (possibly via local institutes or global online like Coursera/IBM or ALX's data program). We could partner with the likes of ALX – which has already trained 85,000+ African learners in tech skills since

2021[18][19] – or Ingressive for Good (I4G) which provided coding training to over 132,000 students[19]. These organizations often have cohorts in Nigeria and could tailor corporate batches for us. Additionally, vendors like Cisco and Microsoft have initiatives (Cisco committed to training network engineers in AI[20]; Microsoft's AI Cloud certifications, etc.). We will seek to "piggyback" on these: e.g. secure some seats in their programs or get volume discounts. - Higher Education & Degrees: For a few key individuals, sponsoring part-time or executive Master's programs (e.g. an M.Sc. in Data Science at a Nigerian university or online university) could be worthwhile for long-term capability. This would be a selective investment (perhaps for future AI team leads). - On-the-job Mentoring & Rotation: Technical learning doesn't stop at coursework. We plan to implement a mentorship where, say, an IT systems engineer who learned Python starts doing real data tasks under an experienced consultant's guidance. We might hire 1–2 expert data scientists (even on contract) who, besides delivering projects, also coach our internal trainees on real projects. *Budget:* This is the most expensive per capita. Bootcamps can cost ₦500k–₦1M per person ($650–1300). If we do 10 people, that's ~$10k. Degrees might cost $5–10k each per year; we might fund a couple. Total over 3 years could be ₦50M ($65k) for deep skilling. While not trivial, this is an investment in building an internal AI Center of Excellence.

Role-Specific Upskilling: In addition to the tiers, we will run tailored training for specific operational roles as new tools come in: - Field Technicians: When we introduce, say, a predictive maintenance mobile app, we'll do hands-on workshops for techs to practice using it. If we deploy drones for tower inspection, we will train a subset of technicians in drone piloting and data interpretation. We may set up a small simulation lab possibly with AR/VR: technicians can practice in a virtual environment (as suggested by industry experts, AR/VR can significantly enhance practical training and reduce risk)[21][22]. This means if a technician needs to learn a complex repair, they could first do it in VR. We foresee partnering with a vendor or university to use AR training kits (for example, TowerXchange reports that AR/VR is helping telecom workforce training globally[22]). - NOC Operators: They will be trained on the unified AI-driven NOC dashboard we plan to implement. This will include scenario drills – e.g. an exercise where multiple alarms come in and the AI prioritizes them; the operator must respond accordingly. Cross-training in basic analytics might be given so they understand things like "an anomaly score" that the AI might present. - Security & Site Monitoring Staff: If we reduce on-site security guards and move to camera/AI surveillance, we must retrain those staff to become remote monitoring operators. They'll learn to use CCTV monitoring software, respond to AI alerts (which might be different from responding to a human call – e.g. an AI might flag unusual motion at a site at night; the staff needs to remotely assess and decide if a dispatch is needed). - Procurement/Finance: As we bring in RPA (Robotic Process Automation) for routine tasks (invoices, expense reports, etc.), these teams will be trained to manage RPA bots – e.g. how to handle exceptions, restart a bot, or tweak a workflow. We'll also train *for new analytical skills*: a finance clerk could grow into a financial data analyst who investigates why an AI flagged a certain transaction as anomalous, etc. - HR Team: Since HR is central to this transformation, we will upskill HR staff on using AI in recruiting, performance management, and employee engagement. For example, recruiters will learn how to work with an AI screening tool ethically – understanding its recommendations but also knowing when to override them to avoid biases (supporting the point that AI + human yields best outcomes, as per SHRM guidance[23]). They will also be trained on data privacy (to comply with NDPR/NDPA in handling candidate/employee data through AI).

On-the-Job Learning & "Digital Champions": Formal training is half the battle – we also plan to cultivate continuous learning: - We will establish a network of "Digital Champions" in each department. These are early adopters or graduates of our training who volunteer (or are selected) to help their peers day-to-day. For example, in a cluster of tower sites, one field engineer who's particularly good with the new app can be the go-to person for others, and we'll give them extra support and recognition. Peer learning is powerful in our culture – many technicians currently learn via apprenticeships, so we're extending that model to digital tools. - Mentoring & Shadowing: Where possible, we will pair staff in transitioning roles with those in new roles. E.g., a current maintenance scheduler (whose job will evolve) might shadow a data analyst in the energy optimization team, gradually taking on tasks like analyzing generator performance data. Over time, that scheduler can become an "Energy Analytics Coordinator." This approach helps transfer knowledge internally. - Knowledge Transfer from New Hires: As we inevitably hire some external experts (data scientists, etc.), we will ensure part of their KPI is to train others. We might formalize this in contract or simply through culture – e.g. requiring them to conduct internal workshops quarterly.

Measuring Progress: We will track key metrics for the training program: - Training completion rates by module (% of target employees who finished). - Assessment scores or proficiency tests post-training (to ensure it's not just attendance but learning). For example, after data literacy training, managers could take a short quiz or practical assignment; our target might be 80% of them score above a certain threshold. If not, we revisit the training content. - Application on the job: through surveys or observation. E.g., 6 months after tool training, are field techs using the app correctly? If not, we do refresher training. This feedback loop is crucial – maybe initial training wasn't effective or too theoretical, so we adapt (perhaps more hands-on practice, or translated materials, etc.). - Internal certification: We might create an internal badge/certification for certain skills (in partnership with HR). E.g. "Certified Data-Informed Manager" if they pass certain criteria. This motivates employees and also signals competence to others.

Budget & Partnerships: Consolidating the rough figures: for an employee base of ~1000 (assumption), we foresee: - Awareness training: largely internal, minimal external cost (<$20k). - Data literacy for ~300 people: ~$75k. - Deep skilling for ~20-30 people over time: ~$75k. - Misc role-specific training (drones, AR, etc.): perhaps $50k allocated, as these will be project-specific (e.g. buying a few AR headsets, contracting a drone trainer). - Total direct training investment ~$200k over 2–3 years (₦150 million). We will refine this with Finance, but it's on the order of ~$200 per employee per year on average. This is about 1–2% of payroll (assuming an average $10k annual salary), which is in line with global best practices for companies undergoing tech transformations (many successful firms invest 1–3% of payroll in training during digital transformation initiatives).

To stretch our budget: - We will leverage free/low-cost programs: e.g. government or donor initiatives. The Federal Government's "Digital Literacy Framework" aims for 95% digital literacy by 2030; agencies like NITDA might offer free training resources. Also, the new Nigeria Startup Act encourages corporate tech training collaborations – we'll explore grants or cost-sharing. - Vendor support: Our tech vendors (tower equipment providers, software vendors) may provide training as part of contracts. E.g., if we buy an AI-powered monitoring system from an OEM, negotiate a training package for our staff. - Economies of scale: Develop a train-the-trainer model internally so we can reuse content. After the first cohort with external trainers, the new champions can teach the next cohort, reducing cost.

Finally, we maintain a no-layoff commitment through this reskilling period. We have communicated (and will keep reinforcing) that roles may change but jobs will be found for those who upskill. For example, if the finance department needs fewer clerks due to automation, we plan ahead: some clerks can be retrained as data entry QA specialists or move into the expanding areas (maybe network operations or sales support if they prefer). Natural attrition (retirements, resignations) will handle some reduction in roles, so we can *redeploy* employees rather than remove them. This commitment is crucial to get buy-in – employees are far more likely to engage wholeheartedly in training if they believe the company is investing in them, not just trying to replace them.

Responsible AI in HR Operations

As we transform our workforce, we will also leverage AI *within* our HR function to support and sustain the change. However, given HR's direct impact on people, we will adopt a "Responsible AI" approach to ensure fairness, transparency, and compliance with Nigerian regulations. This section outlines how AI will be used in HR processes and the safeguards we will implement.

AI Use Cases in HR:

- **Recruitment & Hiring:** We plan to deploy AI tools to improve recruiting efficiency and reduce time-to-hire. For example, an AI-powered recruitment platform can scan CVs and applications to shortlist candidates based on predefined criteria (skills, experience, location), which is valuable when hiring at scale or for new tech roles where we get many applicants. Additionally, AI chatbots can handle initial candidate interactions – answering frequently asked questions about the job or company and even conducting basic screening interviews via chat. Some companies have achieved notable reductions in hiring timelines with such tools[24]. Our goal is to cut average hiring cycle time from ~8 weeks to 6 weeks by Year 2 with AI assistance (a ~25% improvement). This aligns with industry findings – Deloitte reports AI can save about 30% of cost-per-hire by automating stages[25], and 81% of employers using skills-based AI assessments saw lower time-to-hire[26]. That said, we will ensure a human remains in the loop for final decisions. AI will do initial sift and scheduling, but hiring managers and HR will make the ultimate selection, with an eye to diversity and team fit. We will also regularly audit the AI recommendations for bias – e.g. ensuring it's not unfairly favoring or rejecting candidates from certain universities, genders, or ethnic groups. If any skew is detected, we adjust criteria or even disable that feature. Candidate consent and transparency will be maintained: candidates will be informed if an AI is involved in their process and given a way to request human review (in compliance with global best practices and Nigeria's data protection principles).
- **Employee Screening & Internal Mobility:** Similar AI tools can help internally. We can use AI to scan our own employee database (performance records, skill

sets) to identify candidates for internal promotions or new roles. For instance, if we need to staff a new AI unit, an AI could find employees with programming courses or strong Excel skills who might be good fits to retrain. This promotes internal mobility and reduces hiring costs. Of course, we will do this transparently and use it as decision-support. Also, if using personal employee data in AI, NDPR/NDPA requires that employees' data is processed lawfully – likely we will rely on performance data and qualifications which are job-related and allowed, but we'll avoid sensitive personal data.

- Performance Management & Talent Development: AI can analyze performance data to give managers insights. For example, an AI might identify patterns like "Team A has consistently faster site resolution times than Team B" – prompting a best-practice sharing, or flag an individual whose performance dipped significantly compared to last year, prompting a supportive check-in. We'll implement AI in our HRIS to crunch training data, KPIs, perhaps even project outcomes to help spot high-potentials or skill gaps. One idea is an AI-driven personalized learning recommendation: the system suggests courses or assignments to employees based on their role and performance (some advanced HR systems do this[27][28]). We will, however, use these outputs carefully – as inputs to coaching conversations, not as automated decisions on pay or promotion. Any AI-based performance flag must be verified by a manager. This keeps fairness and avoids over-reliance on possibly imperfect data. We will also communicate to employees how these tools work, so they understand that, say, an algorithm might notice they haven't taken training in a while and suggest it, and that this is to help their growth, not "spy" on them.

- HR Chatbots and Employee Self-Service: We will introduce an HR virtual assistant (chatbot) for employees to get instant answers to common HR queries (leave balance, expense policy, etc.). This improves employee experience (24/7 quick help) and frees HR staff from repetitive Q&A. For instance, an employee could ask via Teams chat "How do I apply for maternity leave?" and the bot would respond with the policy snippet and link to the form. We've seen companies reduce HR service waiting times significantly with such chatbots[29]. Ours will be bilingual if needed and tuned to local HR policies. We'll ensure that complex or sensitive queries (e.g. a harassment complaint) are escalated to a human HR officer. Data privacy: the chatbot will be built in compliance with

NDPA – any personal data asked (like "What's my remaining leave?") is accessed securely through our HRIS with proper access control.

- Workforce Planning: AI tools can help HR and operations forecast staffing needs. E.g., by analyzing historical data on network expansion, maintenance demands, and attrition, an AI might project how many technicians we'll need in each region next year. It could also optimize shift schedules or training schedules (perhaps simulating scenarios: "If we send 50 techs for training next month, can the AI find an optimal schedule minimizing impact on operations?" as alluded in our maintenance example[30][31]). We will use such decision-support to ensure we plan recruitment and training proactively. Another area is attendance and overtime analytics – AI might flag if certain teams are regularly overworked, which could risk burnout or safety incidents, so we can intervene (hire more or redistribute load).

- Employee Engagement & Retention: We can deploy AI sentiment analysis on anonymized surveys or feedback to gauge morale. For example, analyzing text from our quarterly "pulse surveys" to identify trending concerns. Also, predictive models could identify employees at flight risk (e.g. based on engagement scores, tenure, career progression speed). If, say, the AI flags that some engineers with 5–7 years tenure are likely to leave (perhaps a common turnover point), HR can proactively roll out retention plans (new career opportunities or benefits for that group). Importantly, as noted in our culture plan, if such a model flags individuals, we use it ethically: as a prompt for supportive action (like a career development talk), never to preemptively push someone out. And we'd be transparent that we use data to improve employee experience (like "we noticed some patterns and want to invest more in career paths around year 5").

Compliance and Ethical Considerations:

Adopting AI in HR raises serious compliance obligations which we will rigorously uphold: - Data Protection (NDPR 2019 and NDPA 2023): Nigeria's NDPA (signed into law June 2023) now provides a comprehensive legal framework for personal data[32]. It supersedes the prior NDPR and establishes the Nigeria Data Protection Commission. Under NDPA, processing of personal data (which HR data certainly is) requires a lawful basis, transparency to the data subject, data minimization, etc. We will conduct a Data Protection Impact Assessment (DPIA) for any AI HR system we implement, as required. For instance, if using an AI to parse CVs, we ensure candidates consent (or it's clearly part of the application process), and that data is stored only as long as needed, protected from breaches, and not transferred abroad without safeguards (if using cloud systems). We'll follow NDPC guidelines and potentially certify our HR processes under the Nigeria Data Protection Regulation Compliance Organisation (DPCO) if needed. - Fairness & Bias: AI algorithms can inadvertently perpetuate bias (e.g. if trained on past hiring data that favored certain schools or genders). We will mitigate this by: (a) choosing AI tools that allow for bias auditing or come pre-trained on diverse datasets; (b) actively excluding sensitive attributes (gender, ethnicity, state of origin, etc.) from consideration; (c) testing outcomes – e.g. run scenario candidates through to see if any pattern of exclusion emerges. We might involve independent auditors or leverage open-source fairness tools. Our Ethics Officer (or committee) will review AI decisions especially in recruitment and promotion to ensure they align with our Diversity & Inclusion goals. - Human-in-the-Loop: A cornerstone of responsible AI is that for decisions significantly affecting people (hiring, firing, promotion), AI should not have the final say. We commit that no HR decision that impacts an individual's employment will be made solely by an algorithm. There will always be a human decision-maker accountable. For example, if an AI screening tool 'rejects' a candidate, we will have HR spot-check a percentage of rejects (especially from underrepresented groups) to confirm nothing valuable was missed. For internal decisions, AI suggestions (like flight risk scores) will remain confidential advice to managers, not labels on employees. - Transparency & Employee Rights: We will be open with our workforce about what AI we use in HR. This includes updating our employee handbook or policies to mention AI-assisted processes. Under NDPA's transparency principle, if employees' personal data is used in an algorithmic decision, they have the right to know. We'll also provide a means to appeal or get explanation for AI-influenced decisions. E.g., if an internal candidate is not shortlisted by an AI tool for a role they applied to, they can ask HR for a review, and HR will manually consider them as well. - Labor Law & Consultation: The Nigerian Labour Act (and potentially union agreements) require fair labor practices. We will ensure none of our AI implementations violate any labor rights – e.g., we won't use AI to monitor employees in a privacy-invasive way (like constant camera surveillance with AI without agreement). Any significant change in how performance is managed or how tasks are allocated via AI will be discussed with employee representatives. We are aware that introducing certain monitoring AI could raise privacy or dignity concerns, so we will tread carefully and only adopt what is clearly beneficial and proportional. - NCC and Industry Regulations: As a telecom infrastructure provider, any tech that could impact network availability (including AI systems controlling aspects of operations) might come under NCC oversight for reliability. We will inform NCC if we implement any AI that materially changes how we manage network uptime, to ensure compliance with any telecom regulations about automated network management. We anticipate no issues, but transparency with regulators builds trust.

By embedding AI responsibly in HR, we are effectively "eating our own dog food" – using in-house what we advocate in operations – which will give us firsthand experience to guide the broader adoption. It will also increase HR's capacity: using AI to automate routine work means our HR team can focus more on strategic partnering with business units during this transformation. HR will thus become a model department in IHS for how AI can augment humans rather than replace them, exemplifying our principle that AI is here to empower our people, not diminish them.

Role-by-Role Impact & "Day-in-the-Life" Scenarios

To make the change concrete, we developed "day-in-the-life" narratives for each major role group, illustrating how work will look after AI integration versus today. We have communicated these in workshops to help employees visualize the future. Here we summarize the expected role impacts:

Leadership/Board – Day in the Life: A future IHS board meeting might include real-time AI-driven dashboards. For example, before a strategic decision, the CEO can pull up a predictive KPI dashboard on a tablet: it shows tower uptime trends, forecasts of energy costs under different scenarios, etc. If a board member asks, "What if diesel prices jump 50%?", management can run an AI simulation on the spot[33][12] and discuss mitigation (e.g. accelerate solar deployments on certain sites). This makes discussions more data-rich. Leadership will spend less time arguing over whose numbers are right (since data is unified and transparent) and more on interpreting what the insights mean for strategy. Another change: decision support via AI in M&A or investments – e.g., evaluating a potential tower acquisition, AI could quickly analyze the target's sites for risks (using historical downtime, local security data). Leaders will need to vet these AI analyses but it accelerates their insight gathering. *Expected personal impact:* Leaders might initially feel overwhelmed by too much data, but we will train them to focus on key signals. They also must model behavior – e.g., using data to justify proposals rather than just "experience". Over time, we expect leadership decisions to be faster and more confidently backed by evidence. There is also a cultural shift: more transparency. For instance, if an AI dashboard highlights a performance issue in a region, it's visible to all in real-time – which encourages a culture of accountability but can be uncomfortable initially (no more hiding issues until end-of-quarter reports). We prepare leaders for that openness.

Middle Management – Day in the Life: Take an Operations Manager for a cluster of towers. Today, they might manually create maintenance schedules, call meetings to allocate diesel, and chase technicians for reports. In the AI future, when they start their day, they log into a system that already generated an optimized maintenance plan for the week, prioritized by AI predictions of which sites are likely to fail soon[13]. Instead of planning from scratch, the manager reviews the AI's suggestions – maybe the AI says "Site X generator likely to fail in 10 days, schedule preventive maintenance in 5 days." The manager checks resource availability and approves. Then they see an alert on their dashboard: "Region A has 10% higher battery usage than Region B." They decide to investigate – this could lead them to coordinate with the Energy Manager or supply chain to check if certain batteries are sub-par. Resource allocation decisions also change: an AI might recommend moving one technician from a low-incident area to a high-incident area to improve overall uptime[34]. The manager's job is to evaluate that – if it makes sense, they justify the transfer with data (which can help in discussions with HR or budgeting). Meanwhile, reporting duties lighten – monthly reports that took days to compile are automated, so managers spend that time coaching their teams or working on improvements. *Challenges:* Some managers might initially feel they've "lost control" because an algorithm is doing what they used to do (like planning). Part of our change management is to show them their role is *still critical*: the AI gives a plan, but the manager's expertise is needed to handle nuances (e.g., knowing a specific client's priority, or an upcoming local event that the AI doesn't know which could affect site access). As they trust the AI more, they'll likely appreciate being freed from drudge work (like less manual Excel) to focus on "managing" in the true sense – solving problems, developing staff, cross-team coordination. We will adjust their KPIs accordingly: e.g., moving from measuring them on how well they followed a schedule to how well they achieved uptime and used AI recommendations effectively.

Field Technicians & Engineers – Day in the Life: Consider a field technician's routine after full AI rollout. In the morning, instead of a fixed daily route, they open their work app which has a dynamic list of tasks prioritized by AI predictions. It might say: "Visit Site 14B – likely generator fuel injection issue. Bring Toolset A and Part #XYZ." The technician drives directly there, guided by the app (which optimized the route). On site, an AI assistant on their rugged tablet or AR glasses can provide guidance: e.g., the tech can say "Show generator maintenance procedure" and get an overlay or video. If they encounter a tricky problem, they might use a support chat where an AI suggests possible fixes based on similar incidents. Drones might have already done a visual inspection earlier that morning (flown by an operator or autonomously) – so the tech already knows tower structure is fine, and they can focus on the generator. After the fix, they log it via voice and take pictures; an AI auto-generates the report. In the afternoon, an urgent alert comes: a site in their area shows an anomaly in battery voltage. The AI marked it as high priority (likely to fail in <24h). The technician gets rerouted immediately (the app shuffles lower priority tasks to next day) to replace the battery before it fails at night – preventing an outage. By evening, because of this predictive approach, they have fewer emergency calls at 2am; their work is more planned and less firefighting. *Benefits:* This means more predictable hours (better work-life balance) and improved safety – since urgent repairs at night or climbing towers in bad weather should decrease[35][36]. We will emphasize to them that AI is reducing the "grunt work" – e.g., traveling to a site only to find nothing wrong, or manually inspecting dozens of bolts where AI could have pointed to the one that's loose. Technicians will do more meaningful technical work – actually fixing things and learning new tech (drones, etc.), which can increase job satisfaction. *Engagement strategy:* We involve some experienced technicians in developing these systems (their know-how is used to train the AI, e.g. we encode their troubleshooting tips into the knowledge base). This inclusion makes them feel valued and that their craftsmanship is still at the heart of operations, just augmented by new tools. We also prepare for initial frustration if AI predictions sometimes err – e.g. if they're sent to a site for a "false alarm". We handle that by continuous tuning and by explaining that every new system has a learning curve (plus we encourage feedback: if they report false alarms, the AI team fixes the model – making them part of the solution).

Network Operations Center (NOC) Operators – Day in the Life: In the future, the NOC will be a more high-tech control room. Instead of manually monitoring thousands of alarms, an AI system filters and correlates them. A day's scenario: the operator sees a screen with maybe 10 prioritized incidents instead of 100 raw alarms. For example, multiple minor alarms (temp sensor spikes, grid outage, battery discharge) might be correlated by AI into "Site X likely on backup power and overheating – high risk of outage in 1 hour." The operator then follows a standard operating procedure (SOP) to remotely address it or dispatch field support. They trust that the AI is watching everything else and will bubble up what needs human action. They also interact with AI-driven automation – e.g., the AI might automatically restart a malfunctioning remote radio unit; the operator just gets a confirmation. *Skill shift:* The operator's skill is in managing the AI – understanding how to interpret its confidence levels, drilling down into raw data when needed, and handling multi-faceted incidents that involve coordination (perhaps network issues plus physical security issues concurrently). They also become testers of the AI – if they notice it prioritizes incorrectly, they flag it to the AI team. *Result:* Operators can handle broader scope (maybe fewer people can manage the same network size, or same people manage a larger network) with lower stress (not glued to screens for every minor blip). We plan cross-training some with security monitoring tasks as well, since both will use similar surveillance AI – a convergence of roles that AI enables.

Back-Office (HR, Finance, Procurement) – Day in the Life: Take an HR administrative officer for example. Today, they might spend time processing leave forms, compiling training attendance, and answering repetitive queries. In the future, many of these tasks are automated or self-service. How does their day look? They might start by checking an AI-generated report of HR metrics: it shows, say, that training completion rates by department, highlights anyone who missed mandatory courses, etc. They act on those insights (send reminder to a manager about their team's compliance, etc.). Next, they could focus on exception cases: maybe the chatbot flagged a question it couldn't answer or a complex leave case (someone taking maternity + annual leave together). The HR officer uses their expertise to handle these special cases – work that is more complex than routine approvals. Similarly, in finance: an AP (accounts payable) staff no longer manually matches invoices; an AI does 90% of them, and the staff deals with the ones that didn't match (maybe a price discrepancy AI can't resolve). Procurement staff might be negotiating with vendors using insights from an AI that tracks vendor delivery performance. Overall, back-office roles become more analytical and supervisory. They'll also engage in project work: for instance, an HR officer might now have time to organize an employee wellness program (something that adds value, whereas before they were swamped with paperwork). *Impact on employees:* Many will likely *welcome* shedding mundane tasks – nobody truly enjoys copying data from form A to system B all day. Studies show automation of drudgery can reduce burnout[37][38]. However, they must adapt to being accountable for AI outcomes. If a payroll bot made an error, employees will call HR – and the HR team must figure out if it was a configuration issue, etc. So we will train them to not blindly trust automation; they need to verify critical outputs (especially early on). They also need to develop a more customer service mindset. With AI handling transactions, the remaining human work often involves directly assisting colleagues or vendors with unique issues –

requiring empathy, communication, and problem-solving. We will include soft-skills training to this effect.

To aid in understanding these changes, we will conduct "Role Impact Workshops": - Each group (technicians, managers, etc.) will attend a session showing old vs new workflows. For example, a *"Day in the life after AI" demo for Field Techs*: we might simulate a typical day with the new system, perhaps with a demo app, to walk them through it[39]. Then discuss their concerns. - Managers might get a simulation game: manage a virtual team with AI inputs (like a management flight simulator). This can highlight how their decision-making shifts from micromanaging to overseeing AI-advised processes. - We incorporate feedback: these sessions are two-way. Maybe technicians have practical suggestions (e.g. "if only the app also did X, it'd help") – we capture those and feed to the development team. - By painting a clear picture of the future, we reduce fear of the unknown. We emphasize how each role's core purpose remains: e.g. "The goal is still to keep towers running and customers happy – that doesn't change, you are still crucial to that; AI is just a new tool to help you."

In summary, *every role* in IHS will evolve, but none become irrelevant. People will spend less time on routine tasks and more on value-adding activities. We will monitor these transitions closely and be ready to adjust – if some employees struggle to adapt to a radically changed role, we will consider repositioning them to roles they can thrive in (perhaps not everyone will take to a highly digital environment, and we can find alternatives in the interim for such cases – e.g. focus them on field work that still needs a human touch, or extend their training and transition period). Our approach is human-centric: augment roles, don't eliminate; clarify changes early; and support people through the journey.

Change Management & Culture Plan

Implementing this transformation is as much about culture and change management as it is about technology. We recognize that introducing AI can provoke anxiety ("Will I lose my job?", "Will I be able to learn this?") and even resistance ("We've always done it this way"). Thus, we have a robust change management strategy that engages stakeholders at all levels, addresses fears, and builds an AI-embracing culture over time. Key components of our plan:

Clear Vision & Narrative: From day one, we need to communicate *why* we are doing this and *what's in it for our employees*. Our narrative: "AI is here to empower our people and help IHS lead the industry." We tie this to our company's mission of enabling connectivity and innovation. For instance, in communications we'll say: *"By using AI to keep our towers running smoothly, we connect more Nigerians reliably – that's our mission. And by learning these new skills, you are keeping IHS at the forefront and securing your future in a tech-driven world."* We also align with company values (teamwork, integrity, etc.): e.g. describe human+AI as the new form of teamwork, or how using data transparently is part of integrity. Having a positive, purpose-driven message helps frame AI not as a threat but as an evolution of our commitment to excellence.

Leadership Example & Sponsorship: Change starts at the top. We will ensure that our Nigeria Managing Director and HR Director (and other execs) are visibly championing the transformation. They will regularly talk about the importance of AI skills in townhalls, share their own learning experiences ("I took an AI for Leaders course – it was eye-opening!"), and model the behaviors we want. For example, regional managers in meetings should start showing data dashboards instead of anecdotal reports, as proof that they trust and use the new tools[40][41]. We'll give leaders talking points and training so they feel confident in this. Also, we might form a Steering Committee for AI Transformation that includes top leaders from HR, Operations, IT and a respected regional manager, etc., which meets monthly to review progress – demonstrating leadership commitment. When employees see their bosses actively engaged and even changing their own habits, it sends a powerful message that "This is real and important."

Two-Way Communication Channels: It's vital to listen as much as inform. We will set up channels for feedback throughout the rollout – e.g., regular "open office hours" (virtually or in person) where project team members address questions, an email hotline for anyone to ask questions or raise concerns anonymously, and pulse surveys to gauge sentiment ("Do you feel you have enough info about the AI initiatives?", "What are you most concerned about?"). This helps catch issues early (maybe a rumor spreading that needs quashing, or a genuine overlooked problem). It also gives employees a sense of ownership – they can contribute ideas. For instance, maybe a field tech suggests a feature for the AI app that we incorporate, which not only improves the project but also increases buy-in among peers ("They listened to one of us!").

Stakeholder Engagement – Unions and Regulators: We will proactively engage with the workers' union representatives (if our workforce is unionized or has a workers' association). Rather than surprising them, we'll bring them into the conversation early, explaining that the goal is to upskill, not downsize, and even seeking their input on training approaches (unions often appreciate being consulted on training and safety matters). We might agree on certain commitments formally, such as no layoffs related to automation for X period, or retraining opportunities for any displaced duties – possibly documented in an MoU. With regulators (like the Federal Ministry of Labour or NCC), we may not need formal approval for internal changes, but we will share our approach in industry forums, positioning it as a positive case study of tech adoption with workforce development. This keeps our reputation positive and might attract support (or at least mitigate any external political risk of "tech replacing jobs" narrative).

No-Forced-Layoffs Policy: We've mentioned it earlier but it's worth emphasizing in change management. Publicly and unequivocally stating that we are not cutting jobs as a result of AI, and backing that up with actions, is perhaps the single biggest factor in reducing resistance. People's first fear is job loss; once that is allayed, they are far more open to change. We'll communicate how we plan to handle efficiency gains: through natural attrition, growth (we expect to expand into new areas, and freed capacity can be moved there), and upskilling. We will also involve employees in planning their own career paths in the new structure (so they see a future for themselves in it).

Incentives and Recognition: To encourage participation, we will introduce incentives. For example, completing key training milestones might be tied to performance appraisals (as a positive factor) or small rewards (certificates, a feature in the company newsletter, etc.). Teams that early-adopt an AI tool and show improvements could receive an award or bonus. We will recognize individuals who champion the change – e.g. an old-school technician who successfully learned the new system and helped others could get a "Digital Champion Award" at our annual staff event. By highlighting success stories ("X team achieved record uptime thanks to smart use of AI – kudos to them!"), we signal that embracing AI is valued for career progression[42][43]. This also appeals to the competitive spirit: other teams will want to get on board to also be seen as high performers.

Training in Change Leadership: We will equip our managers with tools to lead their teams through change. This includes workshops on managing resistance, effective communication, and empathy. We provide a "Manager's Change Toolkit" – talking points for team meetings, FAQs with honest answers they can use when employees ask tough questions ("Will any jobs be lost? What if I fail the training?" etc.), and tips like how to handle an employee who is struggling with new tech. We may simulate some scenarios with them (role-playing a skeptical employee conversation). This prepares middle managers, who are the linchpin – if they are on board and capable, change will percolate; if they are disengaged, their teams will be too.

Employee Involvement & Agency: To overcome fear, we plan to involve employees in the *creation* of solutions. For instance, we might have a "AI Ideas Challenge" where employees can propose small AI projects or improvements in their work. If a field employee suggests a creative use of AI (like using our tower site cameras for something new), we support it and maybe implement it. This transforms attitudes from "AI is being done *to* us" to "We are doing *with* AI." Hackathons or innovation days could be organized (with cross-functional teams, including frontline staff, to brainstorm how to make the most of new tools). Even simple suggestions program can surface great ideas and gives employees a voice. Celebrating these contributions (and implementing the good ones) builds enthusiasm and a sense of control.

Addressing Fears and Building Trust: Despite all positive framing, some fear is natural. We tackle it by being transparent about what the AI systems do and don't do. For example, if we introduce an AI that predicts attrition, we won't keep it secret – that would breed paranoia ("they're watching us!"). Instead, we'd say: *"We have a tool that looks at engagement data to identify if someone might be unhappy. The purpose is to help managers intervene positively – and we do NOT use it to fire anyone. If you're flagged, you'd likely just have a career discussion, which is a good thing."* By openly discussing, we demystify it[44][45]. Also, we emphasize the benefits to employees: safer work (fewer midnight emergency climbs), learning opportunities (making them more marketable in the future), potentially better work-life balance, etc. We'll repeat these messages consistently.

We also plan some fun/engaging activities to build a positive culture around AI: - "AI Day" or Tech Fair: An internal mini-expo where we demo our new AI tools, maybe invite external speakers (from GSMA or a tech firm) to talk about how AI is helping workers elsewhere. Employees can try out AR glasses or see a drone demo. This hands-on exposure reduces fear of the unknown and generates excitement. - Storytelling: Internally share stories of colleagues who have embraced a tool and found it made their job easier. Or how a veteran employee learned to code a simple script at 50 years old, for instance – inspirational anecdotes that "if they can do it, you can too".

- Continuous Learning Culture: We encourage curiosity. Perhaps create internal forums (Teams or Yammer groups) for discussing AI experiences. For example, a group for NOC staff across all regions to share tips on using the new system ("Hey, I found when the AI does X, if we tweak Y it works better."). Peer support communities help normalize the changes and empower users to solve issues collaboratively[46][47].

Monitoring Culture Change: We will actively measure and respond to cultural indicators: - Conduct regular surveys/focus groups specifically on the AI transition (e.g., "Do you feel AI tools make your job easier or harder?", "Do you trust the insights from our new systems?")[48][49]. If certain teams report low trust or morale issues, we do targeted interventions: maybe additional training, or have a respected peer from a successful team come talk to them. - Watch adoption metrics: if we see a certain unit is *not* using a tool that could help them, it could be resistance or lack of training – either way, we engage to find out why. - Feedback loops: e.g. an "AI Transition Council" that includes employee reps from different levels, meeting quarterly to discuss how things are going and suggest tweaks.

Dealing with Persistent Resistance: Inevitably, a small minority may be obstinate ("waiting for AI to fail" mindset[50][51]). Our approach: first, understand them – often resistance comes from fear of change or feeling one's identity threatened ("I've done this for 20 years, I know better than a computer"). We will *engage* them, possibly by giving them a role in improving the system ("Your skepticism is noted – help us test it rigorously to find weaknesses"). By involving critics in the solution, many will soften their stance. For those who simply refuse to adapt even when others have, we may have to handle on a case-by-case basis with HR (through additional counseling, and ultimately, if someone willfully won't perform their role's new requirements, it becomes a performance issue – but we see that as a last resort, hoping our supportive approach prevents reaching that point).

In essence, our change management plan ensures the transformation is not just imposed top-down, but rather becomes a collective journey. We want employees at the end of 36 months to feel *proud* of how far they've come – to see AI as a normal part of how we work (just like computers or mobile phones became normal), and to feel that their company invested in them to make it happen. A strong, adaptive culture will be a lasting competitive advantage: it means that beyond this AI project, the organization will be more agile and change-ready for whatever comes next (be it 5G, robotics, or other innovations). As we often say, "culture eats strategy for breakfast" – so building the right culture around AI is crucial for the strategy to succeed and sustain.

12–36 Month Roadmap (Phased Rollout & Operating Model)

We have structured the transformation into three phases over 36 months, with clear milestones and a governance model (RACI) to manage responsibilities. This phased approach allows for quick wins, iterative learning, and scaling up after proving value in pilots.

Phase 1: Foundation & Quick Wins (0–6 Months)
Objectives: Establish the groundwork – mapping skills, launching initial training, and piloting AI in controlled use-cases to demonstrate value.
- **Month 0–1: Kickoff and Skills Mapping Completion** – Formal launch of the program (announce internally, form project teams). Finalize the detailed skills inventory (as we've done above) and identify individuals for key training (who might be our first data science trainees, etc.). Begin sourcing training partners (issue RFPs or reach out to identified partners like CcHub, ALX, etc.). **RACI:** HR (Responsible for skills assessment), department heads (Accountable to verify role needs), IT (Consulted for tech roles mapping), employees (Informed of upcoming survey/assessment).
- **Month 2: AI Awareness Training Rollout** – Deploy the Tier 1 "AI Awareness for All" e-learning/workshops. Ensure 50%+ staff complete it by end of month. Also run leadership awareness sessions (maybe an exec seminar with an external AI in telecom expert). **RACI:** HR L&D team (R for execution), All Managers (A to ensure their teams attend), IT (C to support any platform), Comms team (I – to amplify messaging).
- **Month 2–3: Pilot Projects Initiation** – Select 1–2 **pilot use cases** of AI in operations and 1 in HR. For example, choose a cluster of towers to pilot a predictive maintenance AI (maybe with a vendor or an in-house prototype), and implement an AI CV-screening tool for one department's hiring. Limit scope so we can manage closely. Develop success criteria (e.g. "pilot reduces emergency repairs by 20% in that cluster" or "AI screening cuts hiring time by 1 week for that dept."). Also, finalize tool selection for these pilots (maybe use an off-the-shelf AI for recruitment to start, rather than building custom). **RACI:** Operations Director (R for ops pilot), IT/AI team (R for tech implementation), Towerco Region Manager where pilot runs (C for ops context), HR Director (R for recruitment pilot), Compliance (C to ensure data privacy from start).
- **Month 4: Policy & Governance Setup** – Draft or update policies: AI Ethics policy, Data Protection compliance checklist for AI, etc. Also create the AI Steering Committee (if not already) and define RACI for ongoing management. Likely structure: HR will lead the workforce transformation program; Operations will co-sponsor since many impacts are there; IT/Technology will provide support for tool implementations; Compliance

will oversee ethics. For example, RACI for an AI implementation might be: IT (Responsible to implement technology), Business Owner (A for ensuring it meets needs), HR (C if it's an HR tool, or I otherwise), Compliance (Consulted for sign-off on data/privacy). We document these and communicate to all involved.
- **Month 4–6: Training Tier 2 launch (Data Literacy)** – Begin rolling out the first cohort of data literacy training for managers/staff (could be 50-100 people in cohort 1). Also send selected tech employees to external bootcamps or online courses as part of Tier 3 (some might start earlier if ready). Implement the digital champions nomination in each department now so they are in training too. **RACI:** HR L&D (R), External training partner (if any, R for content), Department Managers (A to send their people and give them time), IT (I to provide software or data for training).
- **Month 5: Pilot Evaluation & Iteration** – About 2-3 months into pilots, gather data. Did the predictive maintenance AI actually predict any failures? How did techs in the pilot respond? Did the recruitment AI save HR time? We evaluate against criteria. Document lessons and adjust approach: e.g., maybe predictive model needs more data, or technicians need more training to trust the alert. If pilot results are positive, plan scaling; if not, refine and possibly extend pilot. **RACI:** Pilot project leads (R to report), Steering Committee (A to decide next steps), participating employees (C to give feedback), vendor/IT (C/I providing performance metrics).
- **Month 6: Phase 1 Milestones** – By end of 6 months, we aim to have: 100% employees completed AI awareness; at least 50 managers/staff completed data literacy training (or in progress); pilot results showing tangible benefit (even if small scale); and a formal AI governance framework in place. We report these achievements to the board and celebrate internally.

Phase 2: Scaling & Integration (6–18 Months)

Objectives: Broaden the reach of training to most employees, roll out AI tools more widely across functions, and start integrating AI into standard operating procedures (SOPs) and decision processes.
- **Month 7–12: Broaden Training and Hiring** – Continue waves of Tier 2 and Tier 3 training. By month 12, target that all middle managers have gone through data literacy program. Also, by around month 9, we likely see the first batch of internal trainees (deep skill) finishing bootcamps – integrate them into project teams or new roles so they get practical experience. If needed, start hiring external talent for critical gaps (e.g., hire 1-2 experienced data scientists or an AI engineer by month 9) to build the internal Center of Excellence that will support deployments. **RACI:** HR (A for training completion targets and hiring), Department Heads (R to ensure their folks attend, and

for identifying internal candidates for roles), IT (R for onboarding new technical hires, setting up infrastructure for them), Finance (I on budget usage).

- **Month 7–9: Scale Successful Pilots** – If the maintenance AI pilot succeeded, plan rollout to more regions. Possibly purchase necessary licenses/hardware for scale (e.g. install sensors on more towers). Similarly, expand AI recruiting tool to all hiring or additional departments. Develop an *implementation playbook* from pilot lessons to ensure smooth adoption in new areas. Also involve the champions from pilot sites to train others. **RACI:** Operations lead (R to implement in new regions), Regional Managers (A for uptake in their region), IT (R for technical deployment), Finance (C if significant capital spend like sensors), HR (I or R if it's HR tool scaling).
- **Month 9: SOP Integration and Process Redesign** – By now, some AI tools are in use, so update official processes: e.g., update the maintenance SOP to say "System generates schedule, manager reviews…" (formally replacing old manual step). Update job descriptions if needed to reflect new tasks (this may involve HR job evaluation but is crucial for clarity). Develop KPIs that include AI usage, e.g., a manager's KPI could include "Effective use of predictive maintenance system (measured by e.g. % of recommended tasks completed)". **RACI:** Process owners (R to update SOPs), HR (A to update job descriptions/KPIs as needed), Compliance/Quality (C to ensure no regulatory issue with new processes), All employees (I via communications of new procedures).
- **Month 10–12: IT Infrastructure & Data Readiness** – As scale increases, ensure our IT backbone can handle it. This includes possibly setting up a cloud data lake for all tower data, ensuring connectivity at sites for IoT sensors, cybersecurity measures (since more data flowing). Also implement any enterprise AI platforms acquired. Essentially, Phase 2 sees the building of a more permanent AI infrastructure beyond just pilots. We might establish a small AI CoE (Center of Excellence) team formally by end of Year 1, composed of internal experts and overseen by the new Head of AI or similar. **RACI:** IT (R for infra deployment), new AI CoE lead (A for making sure infra aligns with use cases), Procurement (C for any vendor contracts for platforms), InfoSec (C to check security), Operations (I as recipients of the infrastructure's outputs).
- **Month 12 Checkpoint:** At one year, we expect: at least 70% of target staff are trained in their tier (some deep training ongoing into year 2 for advanced roles), at least 2-3 AI use cases operational across the company (e.g., maintenance AI live in multiple regions, an HR chatbot live, etc.), and early signs of benefit (KPIs trending positively). We will do a comprehensive review at this point – possibly involving external auditors or consultants to validate progress – and then refine Phase 2/3 plans accordingly. Present results to board to secure continued buy-in.

- **Month 13–18: Enterprise Rollout and Change Reinforcement** – Complete the roll-out of major AI systems to all intended areas. E.g., by month 18, predictive maintenance AI covers 100% of sites, NOC automation is fully in use, and the HR recruitment AI is used for all roles (with adjustments for exec hiring maybe). This period might involve heavy lifting in training remaining staff on the tools and adjusting any policies (like maybe incentive schemes to ensure people use the tools properly). We'll also likely certify some managers on advanced data skills by this time (maybe requiring passing an internal test or external certification for those who want). Towards the end of Phase 2, we focus on embedding – making the new ways the "new normal". We keep pushing culture: celebrate successes (like "one year of AI – 1,000 hours saved!" event), rotate champions if needed to new areas to support lagging teams. Also, begin looking beyond – maybe identify next wave of AI opportunities. **RACI:** Operations/HR/IT each responsible for their domain's rollout being complete, Steering Committee accountable for overall program staying on track, all employees consulted/informed as appropriate.

Phase 3: Optimization & Institutionalization (18–36 Months)
Objectives: By this phase, most changes are implemented. The focus is on making them sustainable – fine-tuning systems, continuing skill development as technology evolves, and setting up a permanent structure (like an AI Center of Excellence) for ongoing innovation. Also, we integrate this transformation into "business as usual" and explore further innovations (like robotics, etc.) building on the capabilities we gained.

- **Month 18–24: Performance Tuning and Risk Audits** – Now that systems have been live, we do thorough evaluations: Are the AI models performing well (accuracy, ROI)? For example, we might retrain predictive models with the new data collected, to improve their precision. We also audit compliance: conduct an audit that our use of AI in HR has not led to any bias incident or complaint; check NDPA compliance documentation is in order (perhaps even invite NDPC or third-party to review if beneficial). Any issues found, we fix promptly. **RACI:** AI CoE/Data Science team (R for model tuning), Internal Audit/Compliance (R for conducting audits), Dept heads (A to ensure processes are working as intended), Steering Committee (I on major findings, A to approve significant changes).
- **Month 18–24: Continuous Learning & Career Pathways** – By now, new roles have taken shape. We formalize career pathways for employees in the AI-driven org. For example, a technician can see a path to become a "Reliability Engineer" or move into the AI CoE after certain training/experience. We

institutionalize the learning culture: perhaps launch a program like "AI Masters" – ongoing advanced courses accessible to employees, possibly as electives or for new hires as part of onboarding. We might also work with universities to create a pipeline (like offering internships in our AI team to local engineering students, which could turn into hires). **RACI:** HR (R for career path documentation and partnerships), Department mentors (C for shaping realistic paths), Execs (A to endorse and possibly fund education partnerships), Employees (I, and encouraged to partake).

- **Month 24: Embed AI in Strategy & Budget Cycles** – Two years in, we ensure that our annual strategic planning and budgeting fully incorporate AI plans. For example, budgets for 2027 should include line items for maintaining AI systems, continuing training (not one-off, but continuous), and any workforce changes. AI considerations become part of every project: e.g. new tower rollout plan includes a section "AI tools to optimize rollout – costs and benefits". At this point, we might dissolve the special Steering Committee and fold oversight into normal structures (or keep a slim CoE governance). The idea is AI is no longer a separate "project" but part of how we operate. **RACI:** CEO/COO (A to integrate into strategic plan), Finance (R to include in budgets), all departments (C to highlight needs/opportunities).
- **Month 24–36: Center of Excellence (CoE) and Ongoing Improvement** – We expect to formally establish an AI & Analytics Center of Excellence (could be part of IT or a cross-functional team) by around year 2 if not earlier. This CoE will own the AI systems (maintenance, updates), drive new use cases, and support business units with analytics needs. We'll staff it with the talent we groomed + new hires. It might also host regular knowledge-sharing events internally. Meanwhile, Phase 3 is about optimization: fine-tuning processes (maybe some tasks we kept manual we now feel confident to automate by year 3), and exploring expansion. Perhaps by year 3, we consider offering some of our AI capabilities as a service to other companies (a stretch goal – if we develop an especially good system). Or we branch into new technologies like physical automation (drones swarms, robotics for site maintenance) – but those would be separate projects, albeit leveraging our now AI-competent workforce. **RACI:** AI CoE lead (A for driving ongoing improvements), IT (R for systems health), Operations/HR etc. (C to propose new needs), Executives (I on progress, maybe A on any major new initiative funding).

- **Month 30–36: Institutionalize Policies & Wrap-up Project Mode** – Update all HR policies, job descriptions, performance management systems formally to reflect the "new normal" (if any are still provisional). For instance, if new pay scales were needed for new tech roles, finalize those. At 36 months, the transformation project per se concludes – we do a final audit against objectives, document achievements: e.g. "Training delivered to X employees, Y new roles created, key KPIs improved by Z%" (to be reported to board and perhaps publicly as a success story). After this, governance transitions: likely the CoE and normal department management take over, with periodic oversight by a digital transformation committee as part of corporate strategy. **RACI:** HR (R for final policy integration), All dept heads (A for confirming their area is fully transitioned), Board/CEO (I to sign-off that project is completed successfully and becomes BAU).

Gantt-Style Summary of Timeline: (Milestones by quarter) - **Q1 (Months 0–3):** Program launch; baseline training launched; initial AI pilot projects begin. - **Q2 (Months 4–6):** Pilot results evaluated; data literacy training in progress; governance framework active. - **Q3–Q4 (Months 7–12):** Scale up AI tools from pilots; majority of workforce through awareness and many through literacy training; first internal AI experts operational; SOPs updated. - **Q5–Q6 (Months 13–18):** Full rollout of major AI systems; remaining training cohorts completed; culture change well underway (measured by adoption KPIs). - **Q7–Q8 (Months 19–24):** Optimization of systems; audit compliance; permanent CoE/team established; workforce fully in new roles structure. - **Q9–Q12 (Months 25–36):** Continuous improvement; integration into everyday operations; formal close of transformation program with objectives met; preparation for next strategic phase (beyond AI project).

Operating Model & RACI Going Forward:
We define a clear RACI matrix for the ongoing operation of AI-enabled processes and workforce management: - HR Department: *Responsible* for continuing skills development (the internal "AI Academy" to persist for new hires and refreshers), and *Accountable* for overall workforce performance and engagement. HR will also be *Consulted* on any new AI tech introduction (to address training needs and change impact) and *Informed* of performance outcomes to adjust HR strategies. - Operations (Field Ops, NOC, etc.): *Responsible* for using AI tools in daily workflow and hitting operational KPIs with their augmented processes. They are *Accountable* for maintenance outcomes, uptime, etc., regardless of AI – meaning they must ensure AI is effectively utilized or provide feedback to improve it. They will be *Consulted* by IT/CoE

when tweaking AI systems (their feedback on what's working on ground) and *Informed* of any new updates or changes to tools. - IT & AI Center of Excellence: *Responsible* for technology implementation, upkeep, and improvement of AI systems. The AI CoE lead is *Accountable* for the accuracy and reliability of AI models (e.g. if an AI prediction fails and causes downtime, CoE must answer for that and fix it). They're also *Responsible* for data governance (with Compliance). They must work *Consultatively* with HR and Ops (no siloed dev in ivory tower – they exist to serve business needs) and keep all relevant parties *Informed* about new features, outages, etc. - Compliance & Risk Management: *Responsible* for monitoring adherence to regulations (data protection, labor laws, NCC rules). *Accountable* to the board for ensuring no legal breaches or ethical lapses occur. They will be *Consulted* before any major AI deployment or change that could affect compliance (like using employee data in a new way), and *Informed* of ongoing AI decisions and audit results. - Executive Sponsor (could be the COO or CEO for overall program): They remain *Accountable* for the success of the transformation, even after project phase, as part of strategic objectives. They'll ensure resources continue to be allocated. They'll be *Informed* via periodic reports of progress and any roadblocks that need top-level intervention.

This RACI essentially embeds AI responsibilities into normal roles: HR focuses on people, IT on tech, Ops on operations, etc., but now with AI in each domain. The new element is the AI CoE which acts as a bridge. We envision the CoE might eventually be led by a "Head of Data Analytics" reporting to the CIO or COO, with a small team working cross-department.

The end-state operating model is one where AI is integrated in the organizational structure rather than a separate project. We'll have the skills in-house to maintain momentum and adapt as needed beyond the initial 36 months.

KPIs & Expected Benefits (with Scenario Ranges)

To track success and ensure accountability, we have defined key performance indicators (KPIs) across training, adoption, operational performance, and compliance. We also estimate the quantified benefits of this AI-enabled transformation, recognizing uncertainties with scenario ranges (low/base/high). Below are the critical KPIs and benefits, along with how we will measure them and expected targets:

Training & Skills KPIs: - **Training Completion Rate:** % of target employees who complete planned training modules. *Target:* ≥95% for mandatory awareness, ≥80% for tier-2 data literacy within 18 months. We'll track by department monthly. If the low scenario is some resistance or scheduling issues, worst-case we hit ~60% in 18 months (which would trigger corrective action, like making training truly mandatory or integrating into performance reviews). Base case ~80%, high scenario 100% (with strong enforcement and enthusiasm). - Proficiency Levels: Measured via post-training assessments or certification pass rates. E.g., aim that 90% of managers score at least "competent" in a data literacy test by Year 2. For technical roles, aim at least 10 internal staff achieve an industry certification (AWS ML, etc.) by Year 3. - Internal Fill Rate for New Roles: We will monitor the percentage of new AI-centric roles filled by internal candidates vs external hires. We target at least 50% internal by Year 3 (scenario range: low 30% if internal skills lag and we must hire more outsiders; high 70% if our upskilling really pays off). This indicates success of reskilling.

Adoption & Usage KPIs: - **AI Tool Utilization Rate:** For each deployed AI system, percentage of relevant tasks or decisions actually using the AI recommendations. For example, maintenance scheduling – what % of schedules are auto-generated vs manually overridden? We target >75% utilization after full rollout (meaning the AI is trusted and used in majority of cases). Low case maybe 50% if trust issues linger; high could be 90% if it works excellently. - User Satisfaction with Tools: Collected via surveys – e.g., "Do you find the AI tools help you do your job better?" with a scale. Aim >80% positive responses by Year 2. If below 50% (low scenario), then we have either tool issues or change management gaps to address. - Reduction in Manual Work: We'll identify key processes and measure the manual effort before vs after. E.g., HR report prep time per month (target 50% reduction), NOC alarms handled per operator (target 2x increase in ratio due to AI filtering). These are more internal efficiency metrics.

Operational Performance KPIs (enabled by AI): - Time-to-Hire: Average duration from job posting to accepted offer. Baseline perhaps ~8 weeks for technical roles. *Target:* 6 weeks average by Year 2[52]. Low scenario improvement maybe to 7 weeks (if AI helps but bottlenecks remain), high scenario possibly 5 weeks if we streamline aggressively. This KPI ties to recruiting efficiency benefit. - Time-to-Productivity for New Hires: How quickly a new hire reaches a defined productivity level. We expect onboarding improvements with AI (personalized learning, etc.). If currently ~3 months, target to cut to 2 months (33% faster). High scenario could be 50% faster if automated onboarding and training recommendation are highly effective (some studies show up to 50% reduction[53]), low improvement maybe 15%. - Network Uptime / Site Downtime: With predictive maintenance, we aim to reduce unplanned outages. Baseline downtime (e.g. average hours of site downtime per month). Suppose currently 5 hours/site/year; target to reduce by 20% to 4 hours (base case). High scenario maybe 30% reduction, low scenario 10% if adoption issues. This KPI reflects service reliability gains. - Safety Incidents: Count of workplace accidents (especially in field operations). With drones and less emergency rush, target a reduction in incidents, say from X per year to X-30%. High scenario maybe 50% drop in major incidents, as tech removes people from hazardous tasks; low improvement 10% if cultural issues limit this. - Fuel & Energy Cost per Site: AI optimizing generators and batteries can save energy costs. We'll track fuel usage per site. Perhaps target a 10% reduction in diesel consumption in sites with AI energy optimization (scenarios: 5% low, 15% high). This directly ties to cost savings.

HR Process KPIs: - HR Service Delivery Speed: e.g., time to respond to employee HR queries. With chatbots, target 24/7 instant answers for FAQs and reducing average first response from, say, 1 day to immediate for X% of queries. We can measure what fraction of queries the bot handles and employee feedback on resolution time. - Internal Mobility Rate: % of vacancies filled internally. If our upskilling works, this should increase. Baseline maybe 20%, target 30-40% by Year 3. This shows improved skill readiness in-house. - Employee Engagement / Turnover: We will watch attrition rates. A successful transformation ideally improves retention (because employees feel invested in). Our target might be to reduce voluntary attrition from say 10% to <8% annually by Year 3. Low scenario attrition could spike if change poorly handled (>12%), base assume stable or slight improve, high scenario it improves noticeably as people value development.

Compliance KPIs: - NDPA Compliance Audits: Zero major findings in annual data protection compliance audit. (We might do annual self-assessments or use NDPC's framework). Essentially no breaches of personal data from AI usage. - AI Ethics Reviews: We set a KPI that 100% of new AI systems undergo an ethics review before deployment. And zero confirmed cases of AI-driven unfair bias or legal challenge. This is qualitative but important – any incident here would be a red flag.

Now, translating KPIs into quantified benefits (financial or capacity terms): We consider three benefit categories: efficiency (time saved), capacity redeployment (people can do more), and cost avoidance/new value.

Let's quantify some major ones:

- **Reduced Cycle Times / Efficiency Gains:** For example, recruitment time cut by 2 weeks means managers and HR save time. If we have ~100 hires a year, saving 2 weeks per hire is 200 weeks saved; in FTE terms ~4 years of work saved in aggregate (approx 4 FTEs) – they can do other tasks. On the operations side, if predictive maintenance prevents outages, we also save emergency repair time – say each prevented outage saves 5 hours of tech work and avoids revenue loss from downtime. If we prevent 50 outages, that's 250 hours saved. Overall, we estimate perhaps 10-20% productivity improvement in various support functions. Industry benchmarks: companies that effectively train employees can be ~17% more productive[54], and standard onboarding improvements can boost productivity by 50% long-term[55] (though that's optimistic). We conservatively target ~15% productivity gain in targeted areas as base (low 5%, high 25%).

- **Headcount Capacity Redeployment:** Through automation, some tasks equivalent to certain FTEs will be saved. For example, if HR automation saves ~14 hours per week per HR staff (as surveys indicate for manual tasks[56][57]), and we have 10 HR staff, that's 140 hours/week freed (3.5 FTE worth). We won't cut those jobs; instead, those 3.5 FTE capacity can be redirected to value-add work (like employee engagement programs, etc.). Similarly, if NOC automation means we can manage network with, say, 2 fewer operators, we can redeploy them to a growing area (maybe new services monitoring). We will tabulate such *capacity gains*. A key metric: number of roles repurposed vs eliminated. We aim for 0 eliminations, but say 20 roles worth of work is automated by Year 3; those 20 are doing new tasks instead. This is cost avoidance (we didn't have to hire 20 additional people to handle expanded workload) and improved service (because existing staff can focus on quality). We can quantify cost avoidance: 20 roles * average salary (₦ say 5M) = ₦100M/year avoided cost (~$130k). Low scenario maybe only 10 roles worth saved; high maybe 30 if things are very efficient.

- **Operational Cost Savings:** Several areas:

- **Maintenance costs:** fewer failures means less spend on emergency repairs (overtime, express spare parts shipping). Maybe 10% reduction in maintenance OPEX (scenario: if maintenance budget $10M, save $1M, low case 5%, high 15%).
- **Energy costs:** as mentioned, optimize generator runtime and battery use, maybe saving thousands of liters of diesel. If 10% fuel saved on $X fuel budget,

that's direct savings. For instance, if annual diesel spend is $5M, 10% is $500k saving per year.

- **HR costs:** faster hiring can reduce cost-per-hire (less need for external recruiters, etc.). The Deel study we cited says skill-based hiring saved $2,342 per hire[58]. If we fill 100 positions, that's ~$234k saved in hiring costs.
- **Overtime and contractor reduction:** With better scheduling and capacity, we might cut overtime hours (which are paid at premium) or reduce reliance on contract staff. That could be say $200k/year saved if overtime goes down by 20%.
- **Quality & Uptime benefits:** Though harder to monetize, better uptime means we likely hit SLA targets with mobile operators and avoid penalties or unlock performance bonuses. E.g., if in past we occasionally paid penalties for downtime, eliminating those is a saving (say we avoid $100k in penalties).
- **Compliance avoidance:** Ensuring NDPA compliance avoids fines (which can be hefty, up to 2% of annual gross revenue in some regimes). By doing it right, we avoid that risk cost entirely.

Let's illustrate a combined *waterfall of benefits*:

Fig.3: Illustrative annual benefits from AI-enabled workforce (in relative index). Starting at current effort/cost (100), we see reductions from faster cycle-times (e.g. automation and AI analytics speeding up processes saves ~15 units), cost avoidance from not needing extra hires or overtime (~10 units), and efficiency gains by redeploying staff to higher value (effectively another ~5 units saved). Net result is ~30% improvement (final effort ~70) in this scenario. These benefits free up resources for growth rather than direct cuts.

Benefit Component	Impact (Illustrative Index)	Explanation
Current Baseline	100	Current total effort/cost (status quo).
Cycle-Time Improvements	-15	Faster processes (recruitment, maintenance) reduce required effort ~15%.
Cost Avoidance	-10	AI & training avoid extra hires/overtime (~10%).
Redeployment Efficiency	-5	Staff focus on higher-value tasks adds ~5% productivity.

Benefit Component	Impact (Illustrative Index)	Explanation
Future Effort After AI	**70**	~30% total efficiency gain (100 → 70).

In monetary terms (for narrative): if baseline operating cost for relevant areas was, say, ₦1 billion annually, a 30% gain means ₦300 million capacity freed – which we either save or reinvest in other initiatives.

We will develop detailed business case financials as we implement, but initial estimates project a ROI > 150% over 3 years – meaning the gains (cost savings and avoidance) are over 1.5 times the investment cost. Our investment in training and systems (~$0.5–1M total) stands to yield savings on the order of a few million dollars across reduced fuel, improved uptime (retaining revenue), and operational efficiencies.

Scenarios: We maintain three scenarios: - Base Case: Achieve most targets as described. For example, 20% maintenance cost reduction, 25% faster hiring, etc., leading to roughly 30% productivity gain in targeted areas and payback of investment by year 3. - High Adoption/High Benefit: If everything goes very well – strong adoption, AI models exceed expectations – we could see >40% improvement in some processes and additional benefits. For example, perhaps fuel use drops 15%, uptime improves so much that we win additional contracts (new revenue). In this scenario, ROI could be significantly higher, and the company could handle growth with minimal headcount addition, really boosting profitability. - Low Adoption Case: If adoption is slow (some tools not fully used, or benefits less than anticipated), we might only achieve say 10–15% improvement. Maybe cultural resistance means utilization stays at 50% for a long time. In this case, benefits might barely cover costs initially. However, even this scenario can be managed by extending timelines or increasing change efforts. The plan is flexible to adapt if we see lagging indicators – we'd intensify training or simplify tools to drive adoption.

We will continuously monitor these KPIs to ensure we stay on track or adjust course. They will be reported quarterly to the transformation steering committee and semi-annually to the Board. Transparent tracking will also be communicated to employees (e.g., "We as a company have completed 10,000 hours of training – 80% to our goal! And we've cut downtime by 15% so far – great job team!") to motivate and show progress.

Risks & Mitigations

No large transformation is without risks. We have identified the major risks in this AI workforce initiative and developed mitigation strategies for each:

1. **Employee Resistance or Low Adoption:** Employees might distrust AI or be unwilling to change their established routines, leading to under-utilization of new tools (e.g., managers still making manual schedules despite having AI). *Impact:* This would erode the projected benefits and could even cause morale issues if some adopt and others don't.
   Mitigations: We detailed a robust change management plan to preempt this – clear communication (emphasizing no layoffs and personal benefits), involving employees in pilots, and strong leadership advocacy. Additionally, we will monitor usage closely; if certain teams lag, we will intervene with additional training or one-on-one coaching. Peer influence will be leveraged (champions in each group). We also introduced incentives and will incorporate AI adoption into performance evaluations for relevant roles (making it a formal expectation). Essentially, we attack resistance with transparency, support, and if needed, managerial pressure (soft at first, firm if absolutely required). The no-layoff pledge and engagement with unions should significantly reduce active resistance.

2. **Skill Gap not Closing Fast Enough:** It's possible that our training doesn't fully deliver the needed competencies (maybe underestimation of hours needed or difficulty of material), leaving us short on critical skills (e.g., not enough qualified data analysts internally, or managers still not comfortable with data). *Impact:* We might fail to effectively implement AI systems or have to rely more on expensive external hires/consultants, increasing cost and dependency.
   Mitigations: We've partnered with proven training providers and planned hands-on practice to ensure training sticks. We will do periodic skill assessments – if scores are low, we will iterate: perhaps increase training hours, change method (some may learn better with practical projects than classroom). Also, we maintain flexibility to hire externally for roles where internal upskilling is slower (we budgeted for some hires). For critical functions, we might bring in interim specialists to mentor staff (knowledge transfer). We set realistic tiered goals (not everyone needs to become a coder, only select few). If by mid program we see, say, not enough data engineers, we could, for example, outsource certain tasks while doubling down training for another year. The presence of global IHS

resources can help too – maybe borrow an expert from another market short-term. Essentially, we won't hesitate to augment with external expertise while reinforcing internal skilling until it reaches desired levels.

3. **Technology Risk (AI Efficacy and Reliability):** The AI systems might not perform as well as expected – e.g., predictive maintenance AI yields too many false alarms or misses failures, or the HR chatbot mis-answers queries, causing frustration. Also includes downtime or cyber vulnerabilities in new tech. *Impact:* If the tools don't prove reliable, employees will lose trust and abandon them, wasting the investment. It could even disrupt operations (e.g., if an AI wrongly prioritizes and an important alarm is missed).
Mitigations: We will pilot and validate each system carefully before scaling. We involve domain experts in design (to incorporate real-world nuance). We'll start with use-cases where AI is relatively mature (predictive maintenance in telecom has known success elsewhere, we'll use proven solutions). During pilots, we'll run AI in parallel with old method to compare results and only switch over when confidence is high. We also plan thorough testing and iterative tuning (as mentioned in Phase 3, continuous model improvement). For critical systems like NOC, we will keep a human monitoring override until AI consistently proves itself. On reliability: IT will ensure proper infrastructure redundancy; cybersecurity measures will be implemented (e.g., any AI SaaS will be vetted for data security, and we'll comply with NDPC guidelines for data protection). If an AI system underperforms, we have fallback procedures (e.g., revert to manual or simpler automation temporarily) so operations don't suffer. Essentially, cautious deployment and not overhyping AI's capability initially will manage expectations and gradually build trust as the tech earns it.

4. **Data Privacy or Security Breach:** Handling more data (employee and operational) with AI could increase risk of a breach or misuse. A hacker could target our AI systems or an internal misuse could occur (e.g., using employee data improperly). *Impact:* Regulatory fines (NDPA can impose penalties), legal liabilities, and reputation damage. Employees might also lose trust if they feel their data isn't safe or is used in "big brother" ways.
Mitigations: We treat data security seriously – involve our IT security team from the start in all AI projects. We'll enforce access controls: only authorized personnel/AI systems can access sensitive data (principle of least privilege). Data used for AI training will be anonymized where possible (especially if we ever

analyze sensitive HR data, we'll mask identities). Compliance will continuously monitor for NDPA adherence, and we may even pursue NDPC's data protection compliance certification to show we meet standards. Regular security audits of new systems will be conducted, and we'll include AI systems in our existing incident response plan. On the cultural side, we transparently communicate data usage and allow opt-outs when feasible (though for internal operational data, employees generally can't opt-out, we'll still ensure it's only used for legitimate purposes). We also plan to not push certain potential invasive use-cases (like constant employee surveillance metrics) to avoid crossing privacy lines. By staying ahead on compliance (perhaps consulting with NITDA/NDPC), we mitigate legal risk. Insurance coverage for cyber incidents could also be expanded as a fallback.

5. **Cost Overruns or Budget Constraints:** The initiative could run over budget – maybe training costs more if we need extra rounds, or tech solutions cost higher than planned. Alternatively, economic conditions (currency fluctuations, etc.) could constrain funds available. *Impact:* Could force us to scale back or delay parts of the plan, jeopardizing outcomes. For instance, if we can't purchase enough sensors or pay for training all staff, benefits drop.
   Mitigations: We built in some contingencies (using largely internal resources for awareness training, negotiating volume discounts). We'll phase investments – pilot first before big spend, ensuring we only scale what works (avoiding waste). If Naira devaluation or such raises costs, we could prioritize critical components and seek efficiencies (maybe do more training in-house if hiring external became too pricey, etc.). Also, since this program yields cost savings, we will capture and highlight early wins to "fund" later phases (e.g., if first pilot saves $100k in fuel, reinvest those savings into training budget). If needed, we can adjust scope: perhaps extend timeline for lower priority items to spread cost. We will report ROI periodically to keep leadership invested – a strong business case helps ensure budget continuity even in headwinds. Also exploring partnerships or grants (maybe some government scheme to support digital skill training, or vendor co-funding) as earlier noted could provide buffers.

6. **Loss of Key Talent / Change Fatigue:** We are asking a lot of our workforce in terms of change. There's a risk that some high-performing employees might leave due to the uncertainty or because they get poached (with their new skills, they become more marketable). Additionally, the sheer amount of new systems

and training could overwhelm staff, leading to fatigue or burnout. *Impact:* Loss of key people could derail projects (e.g., if a champion or newly upskilled data scientist quits for a higher paying job elsewhere, momentum is lost). Fatigue could reduce productivity and harm morale, counteracting our aims.
Mitigations: Our retention strategy is built-in: we are actively engaging employees, providing growth opportunities, and ensuring they feel valued (this normally increases retention). By being one of the first in our sector in Nigeria to heavily invest in AI skills, we hope to become a magnet rather than a source for poachers – but we must be realistic that skilled data scientists are in demand. We can mitigate flight risk by offering appropriate incentives: possibly retention bonuses for those who complete certain training and stay X years, clear career advancement (so they see staying is worthwhile), and a positive work environment with this new tech (people often leave not just for pay but for better tools and less frustration – we're trying to create exactly that environment). To address change fatigue: we will pace the changes and ensure adequate support. We won't, for instance, roll out five new tools at once to the same group. Also, via our feedback loops, if employees signal overload, we can slow down or provide extra resources (maybe temp staff to handle workload while others are in training, etc.). Celebrating progress and giving people occasional breathers (not scheduling trainings in peak operation periods, etc.) helps. In essence, treat our people humanely and monitor stress – the aim is transformation *with* them, not to burn them out.

7. **External Factors Risk:** Things like regulatory changes, economic downturn, or technology shifts. E.g., government might introduce a new regulation limiting use of AI in certain ways, or power situation might worsen hindering our tech rollouts, or a new AI technology might emerge making our chosen tools obsolete quickly. *Impact:* Could necessitate rework or even halt parts of the plan (for instance, if NCC or Ministry of Labour suddenly put out guidelines restricting AI in HR without human consent beyond what we anticipate, we'd need to adjust processes). Economic issues could mean we can't import equipment or pay vendors on time.
Mitigations: We keep a close watch on the external environment. Our compliance/legal team will maintain dialogue with regulators – perhaps we even participate in industry forums shaping any AI-related policies (so we can anticipate and influence them). We also maintain flexibility in tech choices – favoring modular, cloud-based solutions that we can upgrade or swap if

needed. If a new AI technique comes that's superior, our CoE should be on top of it to incorporate rather than being stuck on an old one (since we invest in continuous learning). For economic issues, we might hedge some costs (like locking in multi-year training contracts in Naira to avoid forex issues). The diverse geographical presence of IHS could also allow some resources from other markets to assist if Nigeria hits a snag (for example, if budgets freeze, perhaps some support can come from Group level temporarily). In the worst case scenario of an economic crisis, we would prioritize sustaining critical operations over new initiatives – but even then, improving efficiency via AI could be part of the solution to do more with less, so we'd argue to maintain core parts of program.

We will maintain a risk register throughout the project, updating likelihood and impact, and assign risk owners to each (for example, HR owns risk of attrition, IT owns tech risk, etc.). Regular risk review meetings (part of Steering Committee agenda) will ensure these mitigations are implemented and adjusted as needed.

In conclusion, while the challenges are non-trivial, our proactive planning and commitment to a human-centric approach significantly reduces these risks. The board can take confidence that we are not naïvely plunging into AI – we are doing so with eyes open, safeguards in place, and a clear focus on empowering our workforce. The benefits in efficiency, cost savings, and competitive positioning outweigh the risks, especially given our careful mitigation strategies. With strong governance and adaptability, IHS Nigeria can navigate this transformation successfully.

Appendices

Appendix A: Data Tables Underlying Visuals

Table A1. Nigeria Mobile Broadband Penetration Data (2015–2024) – source: NCC via TechCabal/Intelpoint[1][3]

Year	Broadband Penetration (%)	Notes
2015	~10% (est.)	Estimated; broadband starting era (4–6% in 2013)[59].
2016	~16% (est.)	Continued 3G rollout, early growth.
2017	19.9%[1]	Sharp increase as data demand rose.

Year	Broadband Penetration (%)	Notes
2018	31.5%[1]	Biggest one-year jump (National Broadband Plan efforts).
2019	~37% (est.)	Approached 40% mark.
2020	45.0%[1]	Peak before SIM-NIN policy impact.
2021	40.9%[3]	Dropped due to SIM registration freeze.
2022	~44% (est.)	Rebound as SIM issues resolved (44.3% by early 2023).
2023	43.7%[6]	Slight decline amid economic factors.
2024	~45.6% (est.)	Modest growth resuming (48.8% by mid-2025)[60].

Table A2. Unemployment Rates – Overall vs Youth (Nigeria)

Year	Overall Unemp. Rate	Youth Unemp. Rate (15–34)	Source / Comments
2018	23.1%[9]	29.7%[10]	NBS Q3 2018 report (pre-COVID economy).
2020	33.3%[2]	~42.5%[2]	NBS Q4 2020 (record highs; youth 15–34).
2022	33%[2]	42.5%[2]	UNFPA 2022 citing youth ~42.5%, overall 33%. (No new official survey; likely similar to 2020).

Table A3. Impact of AI on Hiring & Onboarding (Index 100 = Baseline)

Metric	Baseline (100)	With AI/Automation (Index)	Expected Improvement
Cost-per-Hire (relative)	100	70[25]	~30% cost reduction per hire (AI screening & skill tests)[25].
Time-to-Productivity (new hire)	100	50[53]	~50% faster onboarding (automated, personalized training)[53].

Table A4. Illustrative HR Operations Benefit Breakdown (relative units)

Item	Value (relative)	Explanation
Current HR/Ops Effort/Cost	100	Baseline index.
- Saved via cycle-time reduction	15	E.g. faster hiring, automated reports -> less work.
- Saved via cost avoidance	10	E.g. needed no extra hires due to efficiency.
- Saved via redeployment efficiency	5	E.g. existing staff handle more value tasks.
Future Effort After Changes	**70**	Resulting effort/cost ~70 (30% gain).

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